

Grouped frequency tables and histograms

- 1
- a Ungrouped
- b Grouped

- 2
- a 33.5
- b 21

- 3
- 19 < height ≤ 28

- 4
- a 7
- b 4

- 5
- a

Score (<i>x</i>)	Frequency (<i>f</i>)
10 – 19	3
20 – 29	4
30 – 39	2
40 – 49	1
50 – 59	1
- b

Score (<i>x</i>)	Frequency (<i>f</i>)
30 – 39	5
40 – 49	4
50 – 59	2
60 – 69	8
70 – 79	3
80 – 89	3

- 6
- a 20 → 29, 30 → 39, 40 → 49, 50 → 59, and 60 → 69
- b

Score (<i>x</i>)	20 → 29	30 → 39	40 → 49	50 → 59	60 → 69
Frequency (<i>f</i>)	3	2	1	3	2

7



a $10 \rightarrow 19, 20 \rightarrow 29, 30 \rightarrow 39, 40 \rightarrow 49, 50 \rightarrow 59, 60 \rightarrow 69$, and $70 \rightarrow 79$

b

Score (x)	Frequency (f)
$10 \rightarrow 19$	7
$20 \rightarrow 29$	3
$30 \rightarrow 39$	2
$40 \rightarrow 49$	3
$50 \rightarrow 59$	6
$60 \rightarrow 69$	4
$70 \rightarrow 79$	4

8

a

Score (x)	Frequency (f)
$1 - 5$	6
$6 - 10$	12
$11 - 15$	7
$16 - 20$	3
$21 - 25$	9
$26 - 30$	2
$31 - 35$	4

b

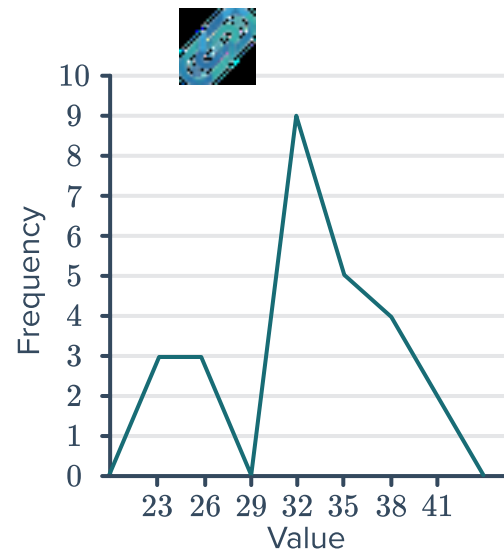
Score (x)	Frequency (f)
$20 - 29$	150
$30 - 39$	140
$40 - 49$	160
$50 - 59$	150
$60 - 69$	130
$70 - 79$	110
$80 - 89$	120
$90 - 99$	130

9

a

Class interval	Class centre	Frequency
$21.5 \leq p < 24.5$	23	3
$24.5 \leq p < 27.5$	26	3
$27.5 \leq p < 30.5$	29	0
$30.5 \leq p < 33.5$	32	9
$33.5 \leq p < 36.5$	35	5
$36.5 \leq p < 39.5$	38	4
$39.5 \leq p < 42.5$	41	2

b



c 0

d 17

e 15

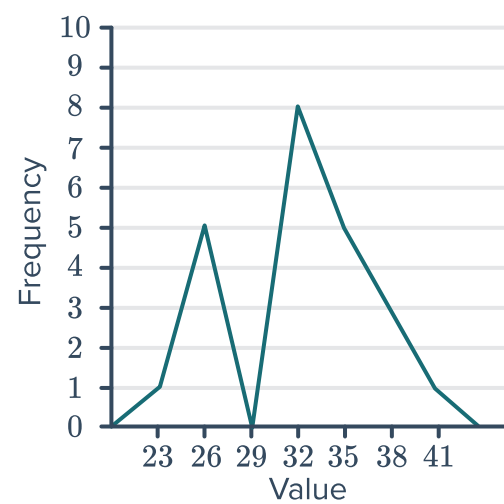
f 20

10

a

Class limits	Midpoint	Frequency
21.5 – 24.5	23	1
24.5 – 27.5	26	5
27.5 – 30.5	29	0
30.5 – 33.5	32	8
33.5 – 36.5	35	5
36.5 – 39.5	38	3
39.5 – 42.5	41	1

b



- c 0
- d 18
- e 14
- f 17



Applications

11

- a Yes
- b Yes
- c Yes
- d No

12

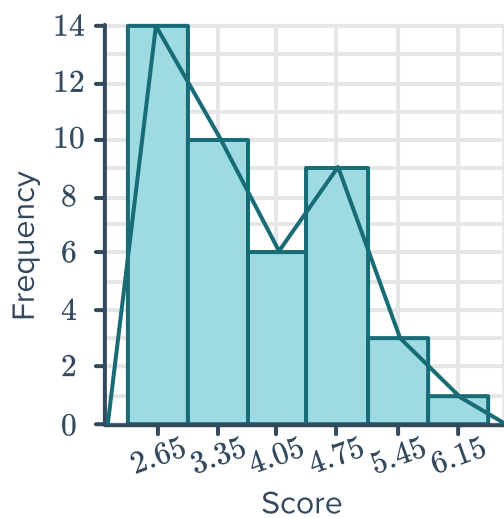
a

Heart Rate	Class Centre	Frequency
45 – 54	49.5	15
55 – 64	59.5	25
65 – 74	69.5	26
75 – 84	79.5	30

- b 96 people
- c 51 people
- d 40 people

13

a



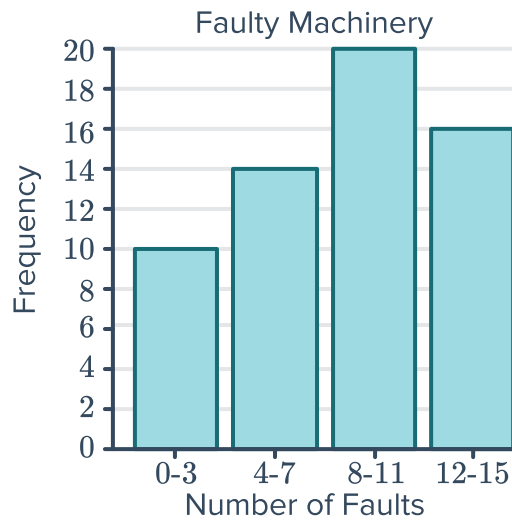
- b The areas are the same.

14

a

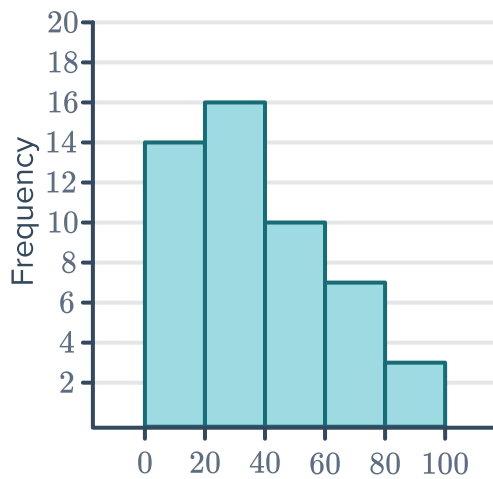


b 0



15

a



b

i No ii Yes iii No iv No

16

a

i $0 \leq \text{rate} < 5$ ii $5 \leq \text{rate} < 10$ iii $10 \leq \text{rate} < 15$ iv $15 \leq \text{rate} < 20$

b 3

17

a 140.9 cents

b 11 days

18

a 25 students

b 13 students

c 16%

19

a

Height (cm)	Frequency
$95 \leq h < 100$	1
$100 \leq h < 105$	3
$105 \leq h < 110$	2
$110 \leq h < 115$	12
$115 \leq h < 120$	19
$120 \leq h < 125$	15
$125 \leq h < 130$	4
$130 \leq h < 135$	4
Total:	60



b 31.7%

20

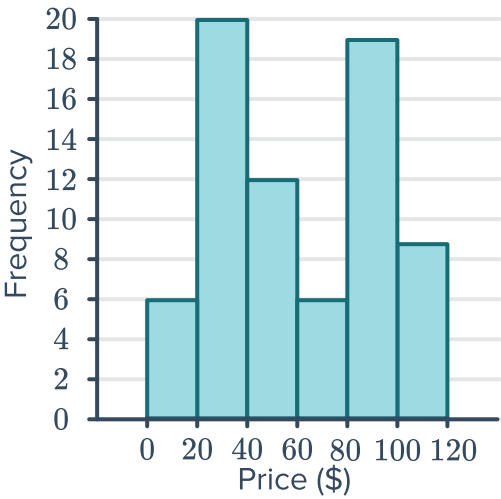
a

Speed (km/h)	Frequency
$110 \leq s < 120$	4
$120 \leq s < 130$	12
$130 \leq s < 140$	8
$140 \leq s < 150$	5
$150 \leq s < 160$	2
Total:	31

b 51.6%

21

a



- c The data shows that there are two main ranges of books that were recently bought by university students: \$20 to \$39, and \$80 to \$99.



22

a

Battery life (minutes)	Frequency
$400 \leq x < 500$	3
$500 \leq x < 600$	9
$600 \leq x < 700$	13
$700 \leq x < 800$	32
$800 \leq x < 900$	22
$900 \leq x < 1000$	27
$1000 \leq x < 1100$	6
$1100 \leq x < 1200$	4
$1200 \leq x < 1300$	1
Total :	117

b 89.7%

c 9.4%

23

a 35

b $60 \text{ mm} \leq x < 80 \text{ mm}$

c Yes

d No. The table shows ranges of lengths and not individual leaves. We do not know if any of the leaves in the range $0 \leq x < 20$ are less than 5 mm.